SUPERSTORE SALES ANALYTICS

Using R Programming with Power BI

Data Visualization Lab • Practical No. 7 • Semester V

Introduction

A sales dataset can tell us much more than just how much a store has sold. By analysing the data properly, we can compare product categories, understand changes in sales over time, and identify areas where the business is making a profit or facing losses.

In this practical, I worked with the **Sample Superstore dataset** to create a Sales Analytics Dashboard using **Microsoft Power BI and R**. The main goal was to bring important sales figures, charts, and interactive features together in one place so that the overall business performance could be understood without having to examine the raw data manually.

This practical also helped me understand how data preparation, visualisation, and statistical analysis can work together to turn raw data into meaningful information.

Tools and Dataset Used

For this practical, I used the following tools:

- **Microsoft Power BI Desktop:** Used to prepare the dataset, create KPI cards and charts, add slicers and filters, and design the dashboard.
- **R programming:** Used for statistical analysis and R-based visualisations.
- **Sample Superstore dataset:** Used to analyse sales, profit, product categories, order dates, regions, discounts, quantities, and other business-related information.

The dataset was suitable for this task because it contained several fields that could be used to compare business performance from different perspectives. For example, sales could be analysed by category, profit could be compared across regions, and monthly trends could be explored over time.

Preparing and Cleaning the Data

Before creating the dashboard, I imported the Sample Superstore dataset into Power BI using the Text/CSV option. I then used Power Query Editor to check the structure of the data and prepare it for analysis.

1. Checking Column Headers

The first step was to check whether the dataset had the correct column headers. Fields such as Order ID, Order Date, Ship Date, Category, Sales, Profit, and Quantity were checked to ensure that the data was organised properly.

2. Checking Data Types

I checked the data types of the columns so that Power BI could interpret the values correctly.

- **Date:** Order Date and Ship Date
- **Decimal Number:** Sales, Profit, and Discount
- **Whole Number:** Quantity
- **Text:** Category, Sub-Category, Region, Segment, State, City, Ship Mode, Customer Name, Customer ID, Product Name, Product ID, Order ID, and Country

I also checked the Postal Code and Row ID columns to ensure that their data types were appropriate for their intended use.

3. Removing Extra Spaces

I checked the Product Name column for unnecessary spaces. Using the Trim option under Transform → Format helped remove extra spaces at the beginning or end of text values.

This step was useful for maintaining consistency in product names and reducing potential issues when grouping or analysing the data.

4. Checking Duplicate Records

I checked the dataset for exact duplicate rows using the Remove Duplicates option in Power Query. The check did not identify any exact duplicate rows, so no records needed to be removed.

Repeated product names or order IDs were retained because they can represent valid transactions rather than duplicate records.

5. Checking Missing Values

I checked important columns, including Order ID, Order Date, Sales, Profit, Category, and Quantity, for missing values. No missing values were found in these fields, so no rows needed to be deleted or values replaced for this check.

6. Applying the Changes

After completing the preparation steps, I selected Close & Apply to load the cleaned dataset into Power BI.

These steps provided a suitable foundation for creating the KPI cards, charts, filters, slicers, drill-down functionality, and R-based visualisations.

Creating the Sales Analytics Dashboard

Once the data was prepared, I started building the dashboard. I arranged the layout so that the main performance indicators appeared at the top, followed by the interactive controls and detailed charts.

The idea was to make the important numbers easy to find while allowing users to explore the data in greater detail whenever required.

Understanding the KPI Cards

The dashboard contains five KPI cards that provide a quick overview of the selected dataset.

- **Total Sales:** 742.00K

```
Total Sales = SUM(SampleSuperstore[Sales])
```



- **Total Profit:** 18.45K

```
Total Profit = SUM('SampleSuperstore'[Profit])
```



- **Total Orders:** 2K

```
Total Orders = DISTINCTCOUNT(SampleSuperstore[Order ID])
```

Total Orders
2K

- **Total Quantity:** 8K

```
Total Quantity = SUM('SampleSuperstore'[Quantity])
```

Total Quantity
8K

- **Profit Margin:** 0.02

```
Profit Margin =
DIVIDE([Total Profit], [Total Sales], 0)
```

Profit Margin
0.02

These values represent the figures displayed in my current dashboard. Total Sales gives an overview of sales revenue, while Total Profit shows the profit recorded in the data. Total Orders and Total Quantity help indicate the volume of business activity.

The Profit Margin card requires particular attention. It currently displays 0.02 as a ratio. If the measure represents profit divided by sales, formatting it as a percentage would make it easier to read. However, the measure and its calculation should be checked before finalising the display.

Exploring the Charts

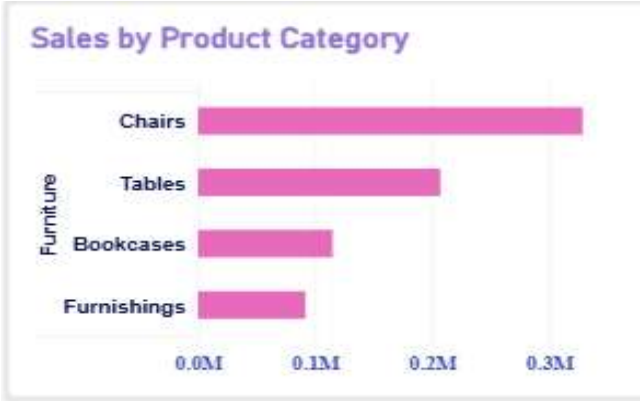
The dashboard contains eight chart areas, divided into two groups. The first four charts were created using the built-in visualisation tools in Power BI, while the remaining four were created

using R visuals integrated into Power BI. This allowed me to explore both Power BI's interactive visualisation features and R's statistical plotting capabilities.

A. Native Power BI Visuals

1. Sales by Product Category

This chart compares sales across product sub-categories. It helps identify which product groups contribute more to overall sales and which have lower sales.



2. Profit by Region — Doughnut Chart

The doughnut chart presents the profit contribution of different regions. It provides a quick visual comparison of regional performance.



3. Monthly Sales Trend

This chart shows how sales change across the months. It helps identify periods with higher or lower sales and explore possible seasonal patterns.



4. Profit by Region — Column Chart

The column chart compares profit across different regions. It makes differences in regional performance easier to identify and helps highlight areas that may need further investigation.



B. R Visuals in Power BI

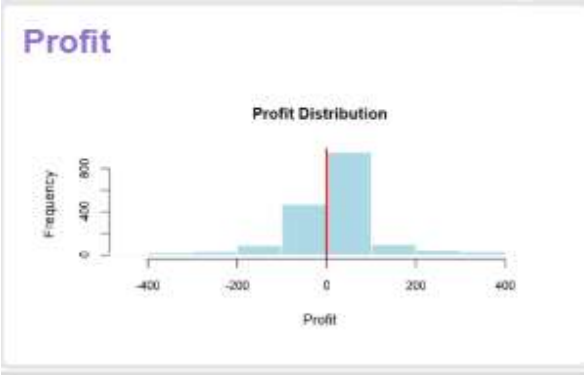
5. Category and Sales

This R visual provides another view of sales across product categories. It helps compare category-level performance using a visualisation created with R.



6. Profit Distribution

This visual explores the distribution of profit values, including values above and below zero. It helps illustrate how profitability varies across the dataset.



7. Discount and Sales

This scatter plot explores the relationship between discounts and sales. A regression line helps show the overall pattern in the data. However, the relationship shown in the chart does not necessarily mean that discounts directly cause changes in sales.



8. Ship Mode and Sales

This R visual compares sales across the available shipping modes. It provides another perspective on how sales are distributed among the different delivery options.

Using both types of visuals helped me understand how native Power BI charts can be combined with R-based visualisations. Power BI provides built-in charting and interactive features, while R offers greater flexibility for customised statistical plots and data exploration.

Together, these eight visuals provide different perspectives on sales, profit, product categories, monthly trends, discounts, and shipping methods.



Making the Dashboard Interactive

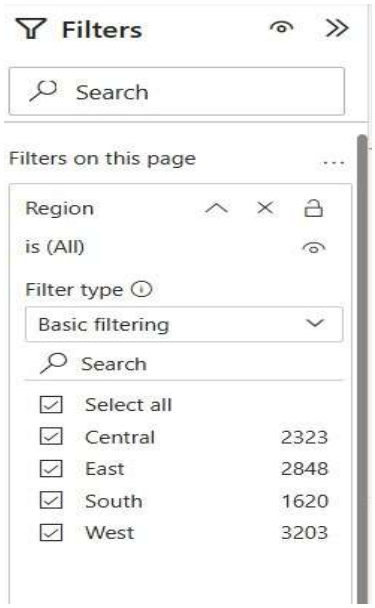
A dashboard becomes more useful when users can explore the data rather than simply view fixed charts. To support this, I worked on page-level filtering and drill-down functionality.

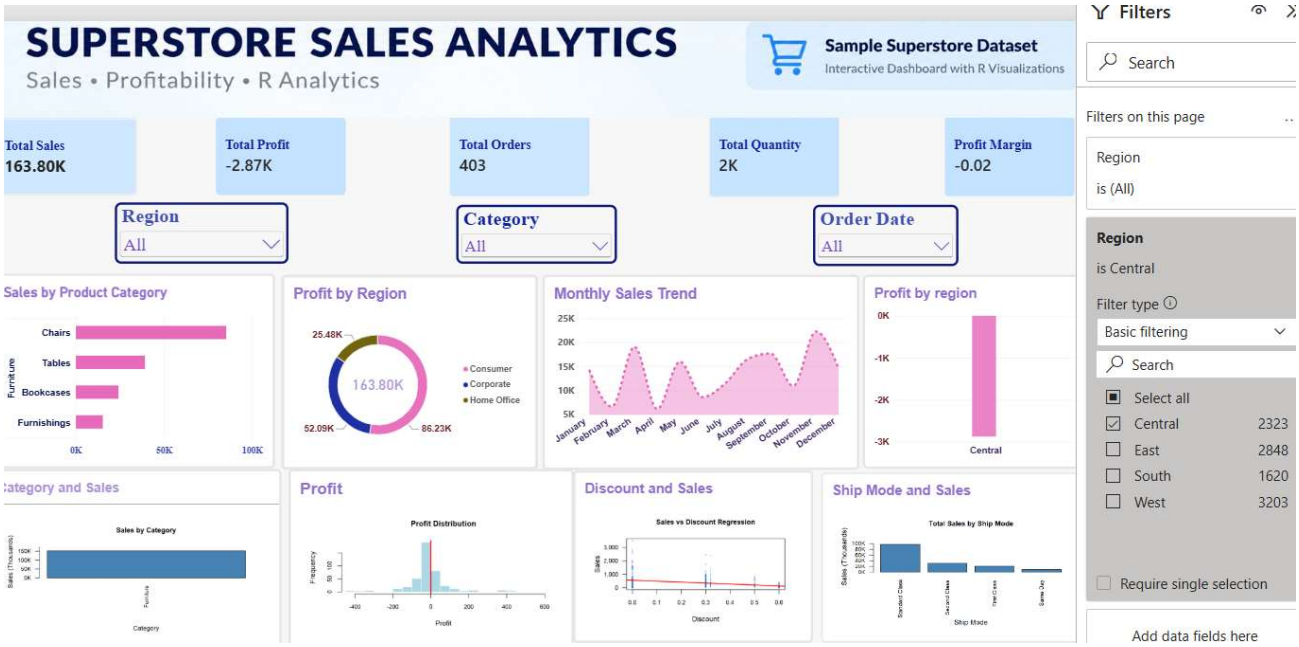
Adding a Page-Level Filter

I added the Region field to the Filters on this page section of the Filters pane.

This allows users to select particular regions and analyse the corresponding sales and profit figures. Clearing the filter restores the view of all regions, subject to any other active filters.

Visual-level filters can also be used when only one chart needs to be filtered. For example, applying a Category filter to an individual visual allows that chart to focus on selected categories without necessarily changing the other charts on the page.





Adding Drill-Down Functionality

I configured a hierarchy for the native Power BI sales chart using the Category and Sub-Category fields.

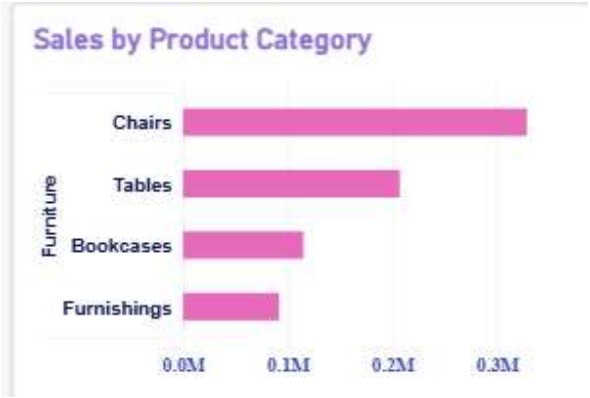
The hierarchy follows this order:

Category → Sub-Category

After enabling the drill-down controls, users can move from a broad category-level view to the individual sub-categories within it. The up-arrow control allows them to return to the higher level.

This feature makes the dashboard more flexible because users can start with an overview and then examine the smaller groups contributing to the displayed results.

The page-level filter and drill-down functionality serve different purposes: filters restrict which data is displayed, while drill-down changes the level of detail shown in a visual.



What I Learned

Working on this practical helped me understand how Power BI and R can be used together for data analysis and visualisation.I learned that preparing the data is an important step before creating any dashboard. Incorrect data types, inconsistent text values, and missing information can affect calculations and the way charts are displayed.

I also gained experience in creating KPI cards, arranging charts, using filters, and building a hierarchy for drill-down analysis. These features showed me how a dashboard can make a large dataset easier to explore.

Another important lesson was that charts need to be interpreted carefully. For example, a region showing lower profit does not immediately explain why its performance is weaker. The next step would be to filter the data, compare the relevant figures, and investigate possible explanations rather than making assumptions based on a single visual.Using R also introduced me to the possibility of extending dashboard analysis with statistical techniques and regression modelling.

Conclusion

The Superstore Sales Analytics Dashboard brings together important sales and profit indicators with charts covering product categories, regional performance, monthly trends, discounts, profit distribution, and shipping modes.

The practical helped me understand the complete workflow, from preparing a dataset in Power Query to creating visualisations and adding interactive features in Power BI. It also demonstrated how R can extend the analysis through statistical methods and customised visualisations.

Before final submission, I need to verify the required number of native Power BI and R visuals, test the slicers and filters, confirm that drill-down works correctly, and include the actual R script output and regression evaluation values if the model is used.

Overall, this practical gave me a better understanding of how raw sales data can be organised and presented in a way that makes business information easier to understand.

Final Dashboard Output

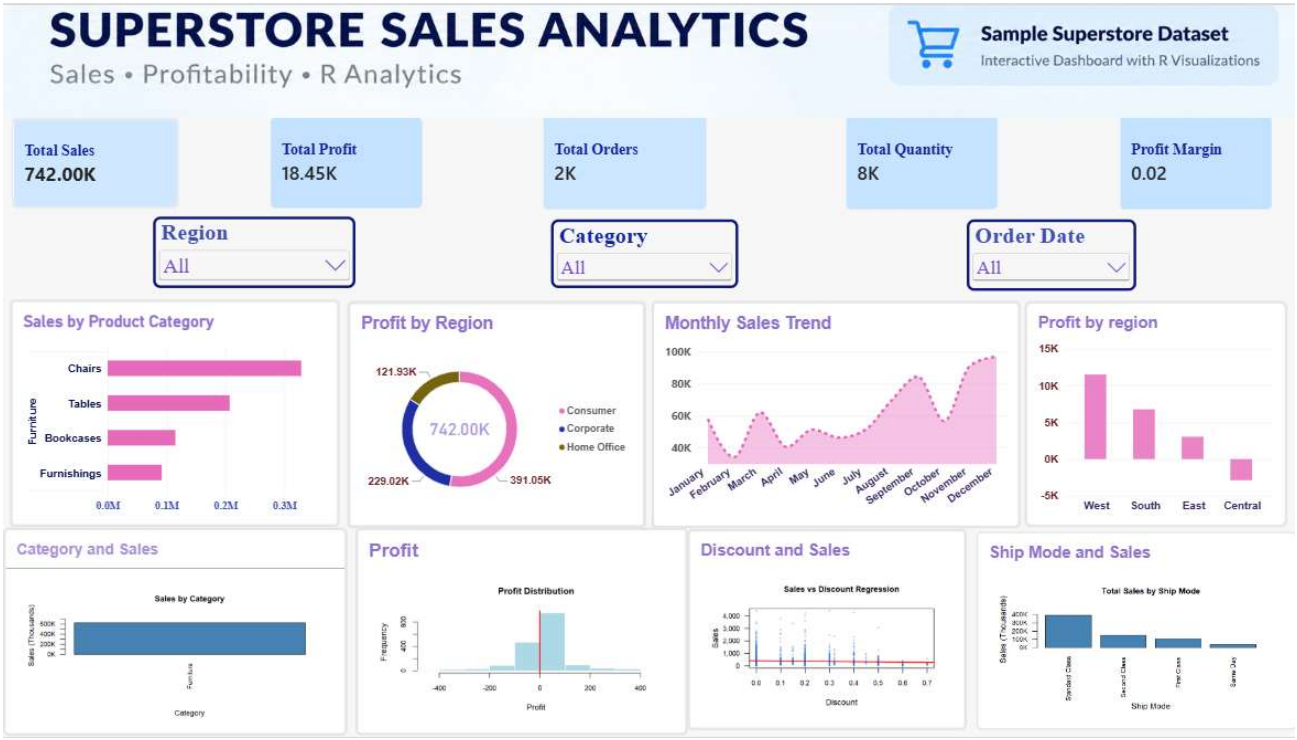


Figure 1. Superstore Sales Analytics Dashboard created in Power BI

Blog Post Link

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